Multiplying Fractions and Mixed Numbers
Answer Key

Grade 4–5

Quick Answers

1. $2 + \frac{2}{3}$
2. $1 \cdot \frac{1}{6}$
3. $1 \cdot \frac{1}{2}$
4. 6
5. 6
6. $1 \cdot \frac{1}{6}$
7. $3 + 1 \cdot \frac{1}{2}$
8. $3 + \frac{3}{4}$ cups
9. $\frac{2}{3}$
10. $6 + \frac{3}{4}$ ft
11. 3.75
12. $7 + \frac{7}{8}$ m^2

Curriculum

Level: Grade 4–5
5.NF.B – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 12 tagged checks.

1. Multiply $4 \times \frac{2}{3}$; write the answer as a mixed number.
Solution: Multiply the whole number by the numerator: $4 \times \frac{2}{3} = \frac{4\times2}{3} = \frac{8}{3}$. Since $8 \div 3 = 2$ remainder 2, that is $\boxed{2\frac{2}{3}}$

2. Multiply: $\frac{1}{2} \times \frac{1}{3}$
Solution: Numerators together, denominators together: $\frac{1\times1}{2\times3} = \boxed{\frac{1}{6}}$

3. Multiply and simplify: $\frac{2}{3} \times \frac{3}{4}$
Solution: $\frac{2\times3}{3\times4} = \frac{6}{12}$, and $\frac{6}{12}$ simplifies (divide top and bottom by 6) to $\boxed{\frac{1}{2}}$

4. Multiply: $\frac{2}{5} \times 15$
Solution: $\frac{2}{5} \times 15 = \frac{2\times15}{5} = \frac{30}{5} = 6$. Answer: $\boxed{6}$

5. Multiply: $1\frac{1}{2} \times 4$
Solution: Rewrite the mixed number first: $1\frac{1}{2} = \frac{3}{2}$. Then $\frac{3}{2} \times 4 = \frac{12}{2} = 6$. Answer: $\boxed{6}$

6. Multiply and simplify: $\frac{3}{8} \times \frac{4}{9}$
Solution: $\frac{3\times4}{8\times9} = \frac{12}{72}$, and dividing top and bottom by 12 gives $\boxed{\frac{1}{6}}$

7. Multiply: $2\frac{1}{3} \times 1\frac{1}{2}$
Solution: Rewrite both: $2\frac{1}{3} = \frac{7}{3}$ and $1\frac{1}{2} = \frac{3}{2}$. Then $\frac{7}{3} \times \frac{3}{2} = \frac{21}{6} = \frac{7}{2}$, which is $\boxed{3\frac{1}{2}}$

8. One batch uses $\frac{3}{4}$ cup of milk. How much milk do 5 batches use?

Solution: $5 \times \frac{3}{4} = \frac{5\times 3}{4} = \frac{15}{4} = 3\frac{3}{4}$. The 5 batches use $\boxed{3\frac{3}{4}\text{ cups}}$

9. Find the missing factor: $\square \times \frac{3}{4} = \frac{1}{2}$

Solution: The missing factor is $\frac{1}{2} \div \frac{3}{4} = \frac{1}{2} \times \frac{4}{3} = \frac{4}{6}$, which simplifies to $\boxed{\frac{2}{3}}$ *Check:* $\frac{2}{3} \times \frac{3}{4} = \frac{6}{12} = \frac{1}{2}$. ✓

10. A shelf is 3 boards, each $2\frac{1}{4}$ ft long. How long is the shelf?

Solution: $2\frac{1}{4} = \frac{9}{4}$, so $3 \times \frac{9}{4} = \frac{27}{4} = 6\frac{3}{4}$. The shelf is $\boxed{6\frac{3}{4}\text{ ft}}$

11. Multiply: $1\frac{1}{2} \times 2\frac{1}{2}$

Solution: $1\frac{1}{2} = \frac{3}{2}$ and $2\frac{1}{2} = \frac{5}{2}$, so the product is $\frac{3}{2} \times \frac{5}{2} = \frac{15}{4}$. Since $15 \div 4 = 3$ remainder 3, that is $3\frac{3}{4}$, or as a decimal $\boxed{3.75}$

12. A garden is $3\frac{1}{2}$ m long and $2\frac{1}{4}$ m wide. What is its area?

Solution: Area = length × width: $3\frac{1}{2} = \frac{7}{2}$ and $2\frac{1}{4} = \frac{9}{4}$, so the area is $\frac{7}{2} \times \frac{9}{4} = \frac{63}{8} = 7\frac{7}{8}$. Area: $\boxed{7\frac{7}{8}\text{ m}^2}$